Auction Course

> Every market is an endless auction between buyers and sellers. This course explains the principle in plain words, walks you through all **13 auction states** of the Flow.42 AuctionEngine, and finishes with how the **AI Agent** turns that into a readable analysis.

At a glance: 13 auction states · Origins in the 1980s (Steidlmayer, CBOT) · 3 building blocks: Price · Time · Volume

1. What is an auction in trading?

Don't picture a chart — picture a marketplace. A market is nothing more than a **continuous two-way auction**: buyers and sellers negotiate until they agree on a fair price — and then the whole game starts over.

> **Everyday picture:** Think of a **flea market**. The seller calls out $50, the buyer offers $20. They haggle until both are happy — say $35. That's exactly what happens in the market, only thousands of times per second. **Price rises to attract sellers** and **falls to attract buyers**. Where both sides are happy to trade, "fair value" is found.

Why price moves at all

Price is an **advertisement**. It moves up or down to attract a counterparty. As long as both sides want to trade at the current price, the market stays calm. When new information arrives (a headline, a large buyer, panic), one side turns aggressive — and price must travel until it finds a new equilibrium.

The three building blocks of every auction:

- **Price** advertises opportunity. A price far from fair value signals a chance and pulls in the other side.
- **Time** validates. The longer the market trades at a price, the more that price becomes "accepted."
- **Volume** measures success. It shows how much actually traded at each price — the most important gauge in electronic markets.

Balance and imbalance — the two states of every market

- ⚖️ **Balance (equilibrium):** Buyers and sellers largely agree on fair value. Price **rotates** quietly in a range and volume forms a **bell curve**. Markets spend most of their time in balance — even though trends leave a stronger impression on memory.
- 🚀 **Imbalance (trend):** New information causes disagreement. One side pushes aggressively, price **trends** and seeks a new value level. A new, shifted area forms where balance will later rebuild.

> **The core idea of "Auction Market Theory":** every up-auction is followed by a down-auction and vice versa. The market forever swings between *seeking balance* and *leaving balance*. Read that rhythm, and you understand the context behind every candle.

Where does this come from? Auction Market Theory traces back to **J. Peter Steidlmayer**, who developed the "Market Profile" at the Chicago Board of Trade (CBOT) in the early 1980s. **Jim Dalton** popularized the ideas in his book *Mind Over Markets*. Flow.42 casts exactly this thinking into a deterministic classifier.

2. The building blocks the engine uses

To understand the 13 states you need a handful of terms. They all come from Auction Market Theory — the AuctionEngine builds them bar by bar from **volume-at-price**.

Term	Meaning
Profile	The distribution of traded volume across all prices of a session. Plotted, it usually forms a bell shape: heavy trade in the middle, thin at the edges.
VPOC / POC	*Volume Point of Control.* The price with the **highest volume** — the session's "magnet," where most participants agreed on value.
Value Area (VA)	The range where roughly **70%** of volume traded (one standard deviation). The session's "fair value" corridor.
VAH / VAL	*Value Area High* and *Value Area Low* — the top and bottom edge of the value area. The most important reference lines of all.
HVN	*High Volume Node.* A price zone with heavy trade → **acceptance**. Acts like support/resistance; price lingers here.
LVN	*Low Volume Node.* A thin zone price **races through** → rejection. Popular for breakout continuation.
Prior vs. Developing	The engine always compares the **completed prior-session profile** (prior: pVPOC, pVAH, pVAL) with the **developing** profile of the current day (developing: dVPOC …).
Acceptance	Price stays outside the old value — confirmed by *time* (enough bars) **and** volume **and** a migrating VPOC. A genuine change of value.
Rejection	Price probes past an edge but finds no takers and returns. The heart of the "failed auction."
POC migration	If the VPOC drifts one way, fair value shifts there → trend. If it stays put → consolidation.
Location	Where is price relative to prior value? (below/at VAL · inside value · near center · at/above VAH). The engine determines this separately from the state.
Confidence	A score of **0–100** the engine attaches to each state. It says how clear the evidence is — *not* a buy/sell signal.

> **Important:** the AuctionEngine produces **no buy/sell signals**. It only supplies *context* — the state of the auction — that the rest of the Flow.42 suite (and you) can use to

> frame your own decisions.

3. The 13 auction states

Bar by bar, the engine assigns the market to exactly one of these 13 states. They follow a clear order of priority: opening first, then failed auctions, then acceptance/imbalance, then balance — and if nothing is clear, "transition."

Color / bias legend: bullish · bearish · caution · neutral

A · Opening states (States 1–4)

Right at the session's start (the first few bars) the engine asks only one thing: **where does price open relative to prior value?** That decides whether a rotation day (open inside value) or a directional day (open outside) is more likely. Acceptance is *not* confirmed yet.

1. Initializing Neutral · `Initializing` The engine has **no complete prior-session profile** yet. Without that reference it can classify nothing. It waits until enough history is loaded. *Confidence 0–20 · Tip: Load more history / Tick Replay*

2. Open Inside Prior Value Neutral · `Open Inside Prior Value` Price opens **inside** yesterday's value area. The market starts where consensus last was — typical of a **rotation day** where price oscillates around fair value. *AMT: "Open-inside" → rotation around VPOC is more likely than a trend. Edges (VAH/VAL) are often tested and rejected. Confidence 60 · Invalidated by: acceptance outside value*

3. Open Above Prior Value Bullish (test) · `Open Above Prior Value` Opening **above** yesterday's VAH. Potentially strong — but still just a **test**, not confirmed strength. It has to show whether the higher price gets accepted. *AMT: Directional day possible. The key is whether the market builds acceptance above pVAH or falls back. Confidence 55 · Invalidated by: back inside prior value*

4. Open Below Prior Value Bearish (test) · `Open Below Prior Value` Mirror of state 3: opening **below** yesterday's VAL. Potentially weak, but at first only a downside test. Treat as a test, not a confirmed downtrend. *AMT: Downside directional day possible, provided acceptance below pVAL follows. Confidence 55 · Invalidated by: back inside prior value*

B · Failed auctions (States 5–6)

In AMT, the **failed auction** is one of the most reliable setups: price breaks past a value edge, finds **no takers**, and tips back into value. The breakout traders are trapped — their stops fuel the reversal. The engine only checks this when the edge was tested *recently* (a memory window).

5. Failed Upper Auction 🔴 Bearish · `Failed Upper Auction` The **upper edge (pVAH) was tested**, but price returned inside value — without convincing upward VPOC migration. The upside breakout has **failed**. *AMT: Classic failed breakout. Don't chase longs up here unless the upper edge is re-accepted. First target: back to the VPOC. Target: pVPOC, then pVAL · Invalidated by: re-acceptance above pVAH*

6. Failed Lower Auction 🟢 Bullish · `Failed Lower Auction` Mirror: the **lower edge (pVAL) was tested**, but price returned inside value — without convincing downward VPOC migration. The downside break has failed, which leans **bullish**. *AMT: Don't chase shorts down here unless the lower edge is re-accepted. First target: back to the VPOC. Target: pVPOC, then pVAH · Invalidated by: re-acceptance below pVAL*

C · Value acceptance (States 7–8)

The opposite of a failed auction. Here the market **accepts** a new value outside the old value area. The engine demands three proofs at once: enough **time** (bars) outside, enough **volume** outside, *and* a genuine **VPOC migration** in that direction.

7. Higher Value Acceptance 🟢 Bullish · `Higher Value Acceptance` Price, time and a migrating developing-VPOC support a **higher fair value**. The market has built value up top and accepted it. *AMT: Context for pullback-continuation — the old VAH can become new support. Not a blind long signal, but a bullish context. Target: prior high / next HVN · Invalidated by: back inside prior value*

8. Lower Value Acceptance 🔴 Bearish · `Lower Value Acceptance` Mirror: price, time and volume support a **lower fair value**. The developing-VPOC migrates down — the market accepts lower prices. *AMT: Context for downside pullback-continuation — the old VAL can become new resistance. Target: prior low / next HVN · Invalidated by: back inside prior value*

D · Imbalance (trend) (States 9–10)

The strongest form of acceptance. On top of value acceptance comes a **range extension** (price stretches the session range clearly beyond the prior extreme) *and* a **strong VPOC migration**. This is a live **trend**, not a mere test.

9. Imbalance Up 🟢 Bullish (trend) · `Imbalance Up` Price accepted **above** prior value, **plus** upside range extension **plus** a VPOC that has clearly migrated up. A clear up-trend context. *AMT: Continuation context. Fades need a clear rejection to work. Target: next upper HVN/LVN or liquidity reference. Highest confidence class · Invalidated by: accepted return into value*

10. Imbalance Down 🔴 Bearish (trend) · `Imbalance Down` Mirror: price accepted **below** prior value, **plus** downside range extension **plus** a VPOC that has clearly migrated down. A clear down-trend context. *AMT: Downside continuation context. Target: next lower HVN/LVN or liquidity reference. Highest confidence class · Invalidated by: accepted return into value*

E · Balance & no edge (States 11–12)

When no breakout is accepted and there's no failed auction, the market sits in **balance**. Here *responsive* behavior (buy/sell the edges) is more likely than trend continuation.

11. Mid-Value / No Edge 🟡 Caution · `Mid-Value / No Edge` Price hugs the **VPOC / center** of value. Trade location is poor: equally far up and down — no asymmetric edge. *AMT: "Protect mode." Don't force trades in the middle. Wait for price to reach an edge, where reward and risk turn asymmetric again. Confidence 78 · Invalidated by: a clean move to a value edge*

12. Balance / Rotational 🟣 Neutral · `Balance / Rotational` The developing value **overlaps heavily** with prior value and the VPOC barely migrates. A balanced, sideways-rotating market. *AMT: Rotation between the value edges and the VPOC. Responsive trading (edges) beats initiative chasing of breakouts that mostly fail. Target: rotate VAL ↔ VPOC ↔ VAH · Invalidated by: accepted break outside value*

F · Transition (State 13)

The catch-all state when the evidence is mixed.

13. Transition / Unclear 🟡 Caution · `Transition / Unclear` The auction evidence is **contradictory**: no confirmed balance, no acceptance, no failed auction. The market is in transition between two states. *AMT: Context is transitional — reduce assumptions. Wait for clear acceptance or rejection before committing. Confidence 45 · Tip: Wait, don't anticipate*

> **How the engine decides:** the 13 states are checked in a fixed order. Opening phase first, then failed auction → acceptance/imbalance → mid-value → balance → and last of all "transition." Each state carries a **confidence (0–100)**, a **location**, an **invalidation point** and a **target**. Everything is computed **deterministically** from time, volume and VPOC migration — traceable and free of randomness.

4. The Auction Analyst — the AI explains the state

The engine tells you *which* state the market is in. The **Auction Analyst** is an optional **AI agent** that, on demand, explains *what that means in context* — in plain language, in your language.

How it works — the human/machine/AI division of labor

1. **Engine** — The deterministic classifier computes state, confidence, location, VPOC/VA — numbers, not opinion.
2. **Snapshot** — On a button click, a structured state snapshot is sent to the Flow.42 backend.
3. **AI Agent** — The model synthesizes and explains the snapshot as a readable analysis — it does not re-classify.

The result appears in its own copyable window (in classic **terminal green** or in white). The AI is deliberately the **explainer**, not the decision-maker: the hard classification stays rule-based and reproducible.

The design principles

- **Engine classifies, AI explains:** The state is computed deterministically. The model only adds the **narrative** — it invents no new signals. That keeps the analysis stable and traceable.
- **Strictly on demand:** Analysis runs **only on a click** — never per bar, never on the render or calculation thread. The chart can never stutter because of it.
- **No API key in the client:** The client stores or sends **no AI API key**. It identifies only with the saved **Flow.42 license ID**; the backend authorizes entitlement and chooses model + system prompt.
- **Failure-safe:** Every error path is defensive: problems are reported in the result window and can **never destabilize** the chart.

In your language

The backend applies the same stable analysis rules across all locales — you only pick the output language: English · Deutsch · Español · Português (BR) · Italiano · Français · 简体中文.

> **In short:** the AuctionEngine is the precise, rule-based eye that reads the auction state. The Auction Analyst is the voice that, on request, translates that state into an understandable story — safe, multilingual, and without ever putting the chart at risk.

Disclaimer: This course is for educational purposes only and is not investment or financial advice. The AuctionEngine produces no buy/sell signals. Trading futures and derivatives carries substantial risk of loss.

Sources on Auction Market Theory

- Tradingriot — Auction Market Theory
- Bookmap — Principles of Auction Market Theory
- FinancialTechWiz — A Trader's Guide to Price Discovery
- Pro Trader Dashboard — Value Area Trading Guide
- Foundations: J. P. Steidlmayer, *CBOT Market Profile* (1985); J. Dalton, *Mind Over Markets*.

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